Richard Larocque

SKILLS

Proficient in C, C++, Ruby, Go
Familiar with Haskell, Python, Java, TypeScript, JavaScript
Experienced at building scalable distributed systems
Deep understanding of Linux
Bilingual (English and French)

WORK EXPERIENCE

Software Engineer — May 2015 — Present
Outreach.io

- As the first engineering hire, helped scale the company from Series A to Unicorn
- Built a durable, ordered, outbox CDC system on top of Rails
- Architected a scalable backend for a real-time activity feed
- Authored readable, blog-style articles to spread knowledge in a diverse team
- Became an in-house expert on Rails, MySQL, PostgreSQL, Elasticsearch, Kafka, and Kubernetes

Software Engineer — August 2011 — August 2014 (Chrome)
Google Inc., Seattle, WA — August 2014 — April 2015 (Infrastructure)

- Helped to maintain and improve Chrome's Sync feature
- Learned to work on software that sees over 100 commits per day
- Devised a sequence CRDT for encoding ordered lists in a NoSQL data store
- Built a pipeline to transform server logs into actionable performance metrics

Design Engineer — May 2009 — July 2011 (full-time)
Avvasi Inc., Waterloo, ON — September 2008 — December 2008 (intern)

- Implemented code to analyze GTP traffic in a live network
- Designed a product licensing solution based on digital signatures
- Improved inter-thread messaging performance with a set of lockless queues
- Helped the company grow from 10 to 50 employees

System Software Engineer — January 2008 — May 2008
Nvidia Corporation, Santa Clara, California

- Developed tests to measure power management performance on Mac OS X
- Helped implement a content protection system based on X.509 certificates
- Worked on the maintenance of Nvidia's Mac OS X drivers

Developer — May 2007 — August 2007
Wind River Systems, Ottawa, Ontario

- Integrated patches into Wind River's Linux kernel
- Reduced Wind River Linux's storage footprint

Linux Software Engineer — September 2006 — December 2006
Netsweeper Inc., Guelph, Ontario

- Implemented efficient user name hashing and lookup for a content-filtering server
- Reverse-engineered a set of TCP protocols used to communicate with hardware firewalls and implemented them in Netsweeper's software
- Wrote a PHP-based web interface for collecting and visualizing SNMP statistics

Computer Scientist — January 2006 — April 2006
Environment Canada, Dorval, Quebec

- Ported an application for analyzing precipitation data from IRIX64 to Linux
- Fixed bugs in a surface model related to a change from 64-bit precision to 32-bit

Model Programmer and Developer May 2005 — August 2005
Environment Canada, Saskatoon, Saskatchewan

- Improved the integration between two numerical weather models
- Built an interactive website to allow hydrologists easy access to the latest weather model output

EDUCATION

Bachelor of Software Engineering Graduated in 2009
University of Waterloo
Waterloo, Ontario, Canada

COURSE PROJECTS

- As part of a team of three, used C++ to implement a compiler for a subset of Java 1.3 targeting x86 assembly
- Wrote an interpreter for a subset of ML supporting Hindley-Milner type inference
- Implemented a one instruction, Turing-complete CPU in Verilog
- Worked with a small team to implement a simple RTOS for the Freescale ColdFire
- Developed a Haskell program to solve the travelling salesman problem and wrote a report comparing its performance and SMP scalability against a competing C implementation

OTHER ACTIVITIES

- See also: `https://github.com/richardlarocque`
- Completed 15 marathons, with personal best of 3:11
- Completed an Ironman Triathlon in 2016